A. DISCUSS BRIEFLY

1. **What is an algorithm, and how is it related to the tasks you perform in your daily life?**
   - An algorithm is a set of step by step processes to do something. We perform algorithms almost everyday when we do something step by step, for example if we are cooking something we first need to have an organized set of tools before we start cooking, or for instance we have to do our laundry, we have to separate the white ones from the colored ones to have it "organized".
2. **Why is it important for IT students to learn algorithms?**
   - Learning algorithms builds problem-solving skills, logic, and computational thinking.

It enables students to write efficient code that scales effectively when handling large datasets or complex processes.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

1. **for (let i = 1; i < 10; i *= 2) { console.log(i); }**
   Step 1: Initialize variable i with a value of 1.
   Step 2**:** Check if i < 10. If it's false, stop the loop.
   Step 3**:** If it's true, output the current value of i (console.log(i)).
   Step 4**:** Multiply i by 2 (i = i * 2).
   Step 5**:** Repeat from Step 2 until i is no longer less than 10.

2. **for (let i = 0; i < 10; i++) { console.log(i); }**
   Step 1**:** Initialize variable i with a value of 0.
   Step 2**:** Check if i < 10. If it's false, stop the loop.
   Step 3**:** If it's true, output the current value of i (console.log(i)).
   Step 4**:** Increment i by 1 (i = i + 1).
   Step 5**:** Repeat from Step 2 until i is no longer less than 10.

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. **What did you observe about the sequence of numbers produced by the first code?**
   - The sequence doubles with each step (exponential progression). The loop skips large gaps between numbers as i grows rapidly.
   OUTPUT: 1, 2, 4, 8
2. **What did you observe about the sequence of numbers produced by the second code?**
   - The sequence increases sequentially by 1 (linear progression), covering every integer from 0 to 9.
   OUTPUT: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
3. **What is the main difference between the way the two loops progress?**
   - First code: Makes use of multiplication (i *= 2), which causes exponential growth and decreases the number of iterations to just four.

- Second code: Makes use of addition (i++), which causes linear growth and makes the program run through each number sequentially (runs ten times).

4. **How does the way an algorithm performs its steps affect its output?**

- The step operator (increment, multiplication, logic branch) is the one that determines the speed at which the program will approach its terminating conditions as well as the particular values it will produce.